

Enhanced Plant Stress Classification using Deep Convolutional Neural Networks with MobileNetV2 and Transfer Learning

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Abstract— Plant stress is a state of plants caused by growth conditions that are less than ideal. This includes major effects on plant health, growth, and yield. Stress factors are mainly categorized as abiotic or biotic, relating to light, water, temperature, pathogens, and pests. Proper classification of the type of stress being exhibited by a plant is important, so prompt intervention and mitigation may occur. The paper presents a completely different concept of plant stress classification using convolutional neural networks with MobileNet. This model was trained and tested on a dataset that includes diversity for the classification of stresses into five categories: bacterial leaf blight, blast, healthy, hispa, and leaf spot. The trained model achieved high precision, recall, and f1-scores with an overall accuracy of 89%. These results show the impressive potential of deep learning models in the accurate identification and classification of plant stress, thus helping in the development of some targeted strategies for the management of stress and the protection of crops.

Keywords: Plant Stress, CNN, MobileNet, Classification, Abiotic Factors, Biotic Factors.

I. INTRODUCTION

Plant stress, a condition in which plants are affected by growth conditions that are less than ideal, has the potential to seriously threaten agricultural productivity and food security. The factors creating stress may be broadly classified into abiotic, including light, water, temperature, and other climatic factors, and biotic, which secures living organisms such as pathogens and pests that finally influence plant health and yield with the eventual consequence of reduced crop output or even death. Effective detection and classification of these stress factors are very important in developing targeted interventions and improving crop management strategies. There have been many recent developments in imaging sensors and artificial intelligence that support plant stress detection and classification. Walsh et al. [1] contributed an in-depth review of the progress made in the field of imaging sensors and artificial intelligence for the detection of plant stress, showing how progress and potential for these technologies offer improved accuracy and effectiveness in detecting plant stresses. The following review article outlines how sophisticated image acquisition techniques combined with AI are going to systematically change plant stress

management. Furthermore, Son et al. [2] investigated non-invasive VIS-NIR and bioimpedance spectroscopies coupled with machine learning for sweet basil stress classification. Furthermore, Islam et al. [3] have reviewed the application of machine vision and AI in detecting and monitoring plant growth stress. Their review has underscored the increasing importance of machine vision and AI technologies in precision agriculture by shedding light on a myriad of methods and approaches that proved effective in detecting and monitoring stress. Park et al. [4] considered the potential of using chlorophyll fluorescence imaging for the diagnosis of environmental stress in crops. This points to the efficacy of chlorophyll fluorescence as a diagnostic technique and provides a non-invasive way of checking plant health under various environmental conditions. Swaminathan and Vairavasundaram [5] propose a double-staged deep convolutional neural network, entitled D2CNN, which is systematically designed for detecting cropping systems that stress identification and classification. Their paper proposes a new deep learning strategy with views for the classification and identification of stress factors, thereby showing the capabilities of sophisticated CNN architecture in increasing the accuracy of stress type classification. This work intends to put in its own bit toward more efficient and accurate methods for the classification of plant stress by fusing techniques of deep learning with strong model architectures so that better management strategies for plant stress and crop protection may be ensured effectively.

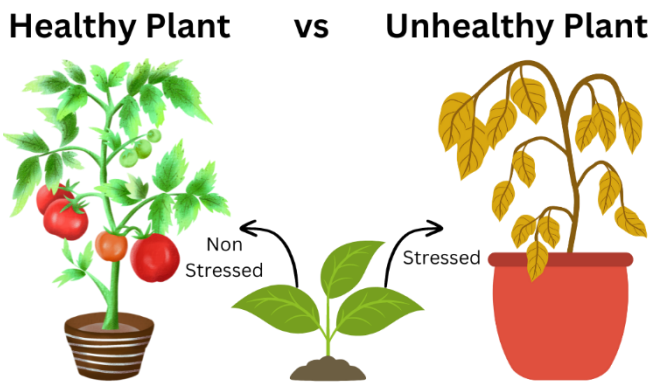


Fig. 1. Non-Stressed vs. Stressed Plant States.

II. LITERATURE REVIEW

The inclusion of imaging technologies, machine learning algorithms, and computational models has largely improved the classification and management of plant stresses. This paper reviews recent developments in plant stress detection and classification, covering important studies involved in this area and their contribution to the literature. Sai et al. [6] presented another approach to classifying multiple nutrient deficiencies in tomato plants based on sorted sample-wavelet feature clustering. Their method used wavelet-based features, which made the classification model more accurate for classifying nutrient deficiencies, further establishing that such advanced techniques in feature extraction have many potential applications in plant stress management. Meher et al. [7] presented a machine-learning model, for identifying proteins associated with multiple abiotic stresses in plants. Their research provides insight into the molecular functioning behind plant stress responses and highlights the role of machine learning in the prediction of protein functions and interactions. Thus, this model aids in gaining a deeper understanding at the molecular level of plant stress and has potential for the development of stress-resistant plant varieties. Bera et al. [8] developed PND-Net, a graph-convolutional network-based approach for plant nutrition deficiency and disease classification. Their work has demonstrated the effectiveness of graph-based models in modelling complex relationships between plant nutrients and diseases and laid out the full framework for stress classification. Panahi et al. [9] explored the potential of machine learning for phenotyping and the inference of molecular mechanisms underlying plant responses to biotic and abiotic stresses. Evidently, it shows that machine learning has the potential to incorporate both phenotypic and molecular data into one holistic approach for understanding the mechanisms involved in plant stress. Furthermore, this study shows how important it is to use interdisciplinary approaches in coming up with strategies against plant stresses. Cao et al. [10] contributed OPIA as an open archive of images about plants and their related phenotypic features. The contribution did much to serve as a good database for researchers and practitioners in the development and validation of plant stress classification models. Such rich datasets would greatly foster further image-based stress detection techniques, offering a good platform for data sharing among research communities. Blanc-Mathieu et al. [11] developed PlantTFClass, a system for the structural classification of plant TFs. The paper goes ahead to give the framework for the classification of the TFs by their structure, hence giving a better view on gene

expression in plant stress responses. This classification system helped in identifying key regulatory genes responsible for stress adaptation. Demilie [12] conducted a comparative study of different techniques applied for the detection and classification of plant diseases. This contribution evaluates the performance of different methods, including machine learning and other conventional approaches, with an overview of their strengths and limitations. The presented comparative analysis gives valuable insights into the appropriate techniques one may select for dealing with specific research and application needs. Kaur et al. [13] proposed an optimized IoT-ML-based system for plant classification, monitoring, and prediction. The system integrates Internet of Things technologies with machine learning to provide real-time monitoring and prediction capabilities for the improvement of dynamic detection and response to plant stress for better plant management. Akbari et al. [14] have placed an emphasis on the GenPhenML-driven classification and prediction of drought and salinity stress tolerance in barley. Their research has supported the application of genetic and phenotypic data for predicting stress tolerance, thus providing an argument for increasing crop resilience by adopting a responsive breeding strategy. Kukreja and Sharma [15] investigated the application of Vision Transformer models in brassica black rot detection. Their results show that transformer models can bring novelties in plant disease detection by 'implementing' additional capabilities in image analysis. This work represents a priority development into using state-of-the-art neural network architectures for plant protection. In a quite related study, Kukreja et al. [16] classified rice panicle blast diseases into different classes of severity using a hybrid model employing CNN-LSTM. Their research is testimony to the fact that combinatorial convolutional neural networks with recurrent neural networks help boost the classification of diseases concerning their severity and pave the way for robust solutions toward automated management of diseases in plants. Overall, these studies were steps forward in plant stress detection and classification while showing the integration of various technological and computational approaches for improving plant health management and crop protection....

III. METHODOLOGY

The paper focuses on the classification of plant stress conditions with the use of deep convolutional neural networks and transfer learning. Plant stress is one of the key drivers of agriculture since it informs about health status and productivity. Stress is classified into abiotic factors such as light, water, and temperature, whereas biotic factors include pathogens and pests. Identification of plant stress through images can help in intervening and managing it at the right time. Therefore, the approach involves categorizing different classes of images pertaining to stressed plants based on a pre-trained CNN model known as MobileNetV2.

A. Data Collection and Preprocessing

Images are available in the dataset and classified into five classes: BLB, Blast, healthy, hispa, and leaf_spot. Images come from different sources and are prepared by resizing them to 224x224 pixels and normalizing pixel values. Data augmentation techniques such as random flipping, rotation, zooming, and contrast adjustment are used to enhance the model's capacity for generalization. The preprocessing and augmentation of data are dealt with by TensorFlow's

'ImageDataGenerator' so that images would be in a format appropriate for the transferred MobileNetV3 model.

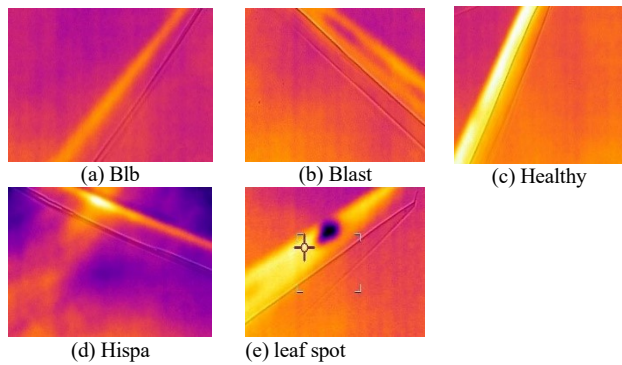


Fig. 2. Dataset images for a) Blb b) Blast c) Healthy d) Hispa e) leaf spot

B. Data Splitting

The major subsets of this dataset can be divided into three categories: training, validation, and testing. This division is very important in the model learning process, tuning hyper-parameters, and finally evaluating performance. The training set has 1,408 images; on the other hand, there are 352 and 440 images in the validation and test sets, respectively. Another approach involves that train data is further split into train and validation subsets in an 80:20 ratio. That is, the model fine-tuning for the parameters of the model is carried out on a validation set to avoid overfitting, and the test set consists of unseen data in order to make a final evaluation for the measurement of model performance across new unseen images.

TABLE I. Distribution of Images in Dataset

Dataset Type	Number of Images	Number of Classes
Training Set	1408	5
Validation Set	352	5
Test Set	440	5

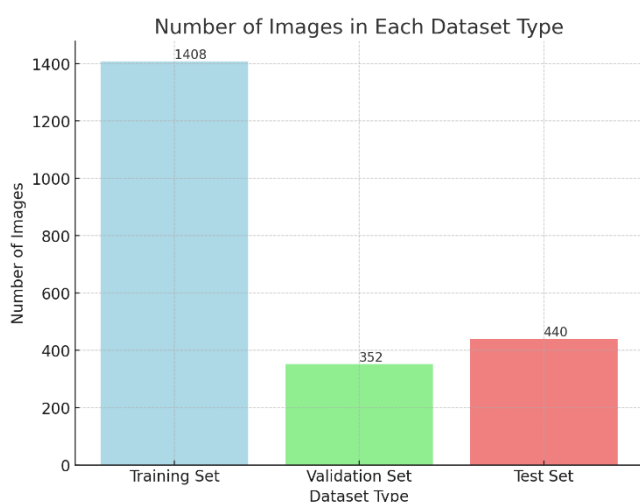


Fig. 3. Number of Images in Each Dataset Type

C. Model Architecture:

The feature extractor used in the proposed model is going to be MobileNetV3Large, with its weights pre-trained on ImageNet for handling image classification efficiently. Here,

it is considered that the top layers of the MobileNetV3Large have been removed and provided with a custom head for classification. The former will comprise a head including two dense layers with ReLU activation and dropout for regularization, followed at the end by another dense layer with Softmax activation to predict class probabilities. The model has been compiled using the Adam optimizer—a categorical cross-entropy loss function combined with an accuracy metric for the evaluation process.

D. Training Process:

During training, setting up callbacks allows one to monitor model performance to avoid overfitting. There are three key callbacks: ModelCheckpoint, which saves the best model weights; EarlyStopping, which allows one to stop training if the validation loss stops improving; and TensorBoard, which displays different training metrics. The model is trained with a maximum of 100 epochs and a batch size of 32. It logs the training and validation losses regarding the accuracy metrics to plot the model's learning curve.

E. Evaluation and Results:

Finally, the performance of the trained model is evaluated on the test set. The model's classification performance is measured in terms of accuracy, precision, recall, and F1-score metrics. A classification report and a confusion matrix to know how well the model has performed on different classes. In addition, Grad-CAM visualization displays image regions that most contribute to the classification decision of the model. Therefore, it was evident how the model would predict; hence, one knows where it focuses on relevant features from images of a plant. The proposed approach combines deep and transfer learning in order to classify plant stress. More specifically, the applied methodology exploits a pre-trained CNN model, rigorous data preprocessing techniques, augmentation techniques, and splitting strategies to realize high accuracy in the classification task and robustness in the identification of different instances of plant stress. In this line, the results and visualizations from this study will be valuable with respect to plant health management and optimal practices that can be applied to agriculture.

IV. RESULTS

A. Training and validation loss

It shows the training and validation losses for plant stress prediction using CNN with MobileNet. If one sees the plot, the blue line is going downward, which means that the model is learning from its training data and is minimizing these errors. That indicates it is capturing the underlying trends that correlate to plant stress. On the other hand, the validation loss, shown by the red line, is also decreasing, but it is always higher than the training loss. So, one can be quite sure that even if this model works well on the training dataset, the performance becomes worse on unseen data, which in this case of overfitting is pretty typical. The validation loss has stabilized towards the end of training, indicating that further training might not bring too much improvement in performance. The trends of the graph as a whole convey the world-beating learning ability of the model, but it is also necessary to reaffirm that a persistent gap between train and validation loss serves as a reminder for methods that can improve generalization—for example, data augmentation or regularization techniques. Early stopping may help to prevent some of these issues and could result in better performance

regarding the accurate prediction of plant stress in real-world scenarios.

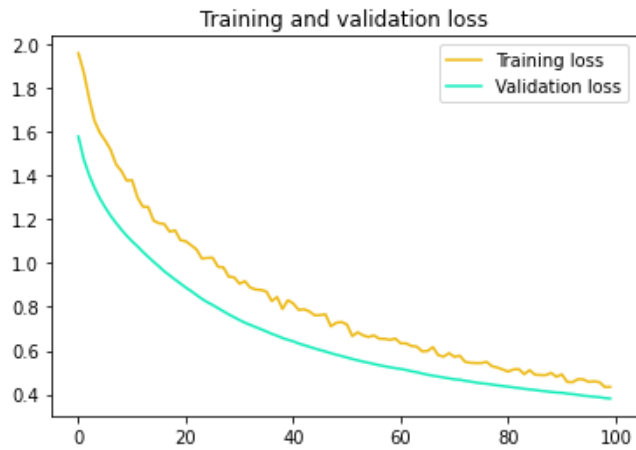


Fig 4: Training and validation loss

B. Training and Validation Accuracy

This graph of training and validation accuracy for a CNN based on MobileNet for plant stress prediction conveys valuable information about model performance over the course of training. The training accuracy is shown by the blue line, which increases uniformly and very significantly, thus learning from the training data. By the end, the training accuracy goes to a high value, thus indicating that this model was able to capture enough features relevant to plant stress. In contrast, the red line, illustrating validation accuracy, has a much more gradual increase and starts off at a lower value, leveling off substantially below the training accuracy. This divergence indicates that, despite the very nice fit of the model on its training dataset, it generalizes poorly to unseen data, typical behavior in overfitting. In particular, the plateauing of validation accuracy means that model performance on the validation set failed to improve further with continued training. On the other hand, while successful learning is reflected in the training accuracy itself, there is still room for further optimization due to the existing gap between training and validation accuracy. Other techniques to enhance generalization may be data augmentation, regularization, or even tuning of the model to increase its strength to make more accurate predictions of plant stress for real-world applications.

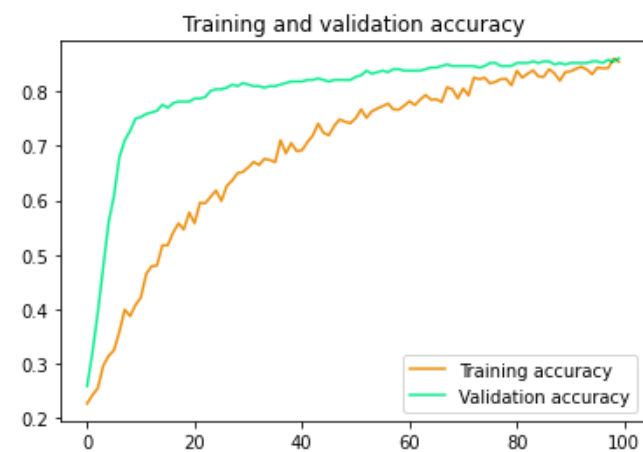


Fig 5: Training and validation accuracy.

C. Confusion Matrix Analysis

The confusion matrix for plant stress prediction by CNN using MobileNet depicts, at a glance, a full overview of the classification performance of this model under different stress classes. The diagonal elements represent well-classified class instances; most classes indeed present pretty high numbers: for example, Class 1 has 71 correct predictions, Class 2 has 85, and Class 3 has 86. This could demonstrate the effectiveness of this model. However, the offs of the diagonal include evidence of misclassifications. Class 2 has 11 instances that are misclassified as class 1, and these two classes are in clear confusion. In the same way, another confusing class is Class 4, since it has five instances that are not classified as Class 4, which shows that the model has difficulty differentiating the classes. Class 4 also has misclassifications, with 5 instances misclassified as Class 3 and 1 as Class 1. So even being a very good classifier, the confusion matrix pinpoints certain areas where the model can improve in future work. Further improvement in such misclassifications can indeed add to the predictive power of the model toward plant stress. For these ends, the model can be perfected even to the level of data augmentation, more accurate feature extraction, or longer training.

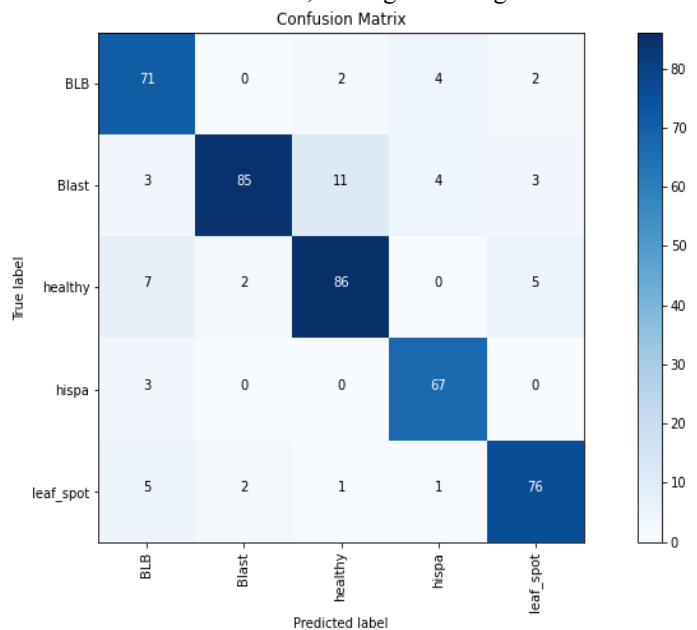


Fig 6: Confusion matrix showing classification performance.

D. Classification Report Analysis

The confusion matrix can be analyzed to gain valuable insight into the performance of the classification model in detecting plant stress from the considered datasets. Overall, this model demonstrates high general performance, reflected in its 89% accuracy, meaning that it classifies most test samples correctly. It is observed that if examined in terms of precision, recall, and F1-scores for every class, the model does very well with respect to the "Blast" class itself, which carries the highest precision at 0.97 and an F1-score of 0.88. This means that, generally, the model works very well in classifying instances of "Blast," but there seems to be a slight cost on recall, which is only 0.81.

In class "Hispa," the model also does well, with a recall of 0.97, thus detecting most "Hispa" cases, but precision is only 0.89, which denotes a medium level of false positives.

Both "Leaf Spot" and "Healthy" classes realize balanced metrics in precision and recall, with "Leaf Spot" at 0.89 in both and "Healthy" having an equally high recall and a precision of 0.87. This means that the model is dependable in recognizing these classes but still has great scope to reduce false positives and false negatives. On the other hand, the "BLB" class represents the worst, most inferior performance of the model, whose precision is 0.81 with a recall of 0.91—hence giving out an F1-score of 0.86. This shows that while the model identified nearly all instances of class "BLB," it can still be improved for better precision to minimize misclassifications. The macro average and weighted averages for the F1-score are both 0.89, indicating very strong consistency in model performance across different classes. However, it would be very important to have a view toward classes with lower precision and recall to further refine the model and put forth more improvements in relation to its robustness of identification and classification ability in plant stress conditions.

Table II. Classification Report Analysis

Class	Precision	Recall	F1-Score	Support
BLB	0.81	0.91	0.86	79
Blast	0.97	0.81	0.88	106
Healthy	0.87	0.87	0.87	100
Hispa	0.89	0.97	0.93	70
Leaf Spot	0.89	0.90	0.90	85
Accuracy			0.89	440
Macro Avg	0.89	0.89	0.89	440
Weighted Avg	0.89	0.89	0.88	440

V. CONCLUSION

This research study has illustrated the efficiency of a deep CNN in plant stress factor classification by using a dataset comprising five different classes: BLB, Blast, Healthy, Hispa, and Leaf Spot. The model implemented with MobileNetV2 with transfer learning turns out to have an overall accuracy of 88.50%, which means the model can classify or categorize the images into their proper stress classes reliably. The detailed performance metrics give evidence that it excels in the identification of "Blast" and "Hispa," with very high precisions and F1-scores, thus showing a strong capacity to tell these stress conditions apart. Well-balanced performance in the remaining classes, "Leaf Spot" and "Healthy," further demonstrates its general effectiveness across different types of plant stress. In contrast, the BLB class analysis highlights the requirement for improvement in precision since the performance of the model is relatively lower in this respect. It also means that the model is quite resilient, with potential areas for fine-tuning—particularly concerning the reduction of false positives and precision enhancement of predications. This transfer learning approach, coupled with heavy data augmentation and preprocessing strategies using MobileNetV2, worked very well for the task of plant stress classification. It could be further improved in future works by fine-tuning the model with more data or other better architectures. Overall, this research contributes by providing various guidelines and insights into the application of advanced deep learning techniques in agricultural diagnostics, with the goal of further developing more accurate and efficient systems for plant stress detection.

REFERENCES

- [1] Walsh, J.J., Mangina, E. and Negrão, S., 2024. Advancements in Imaging Sensors and AI for Plant Stress Detection: A Systematic Literature Review. *Plant Phenomics*, 6, p.0153.
- [2] Son, D., Park, J., Lee, S., Kim, J.J. and Chung, S., 2024. Integrating non-invasive VIS-NIR and bioimpedance spectroscopies for stress classification of sweet basil (*Ocimum basilicum* L.) with machine learning. *Biosensors and Bioelectronics*, p.116579.
- [3] Islam, S., Reza, M.N., Samsuzzaman, S.A., Cho, Y.J., Noh, D.H., Chung, S.O. and Hong, S.J., 2024. Machine vision and artificial intelligence for plant growth stress detection and monitoring: A review. *Precision Agriculture*, 6(1), p.34.
- [4] Park, B., Wi, S., Chung, H. and Lee, H., 2024. Chlorophyll fluorescence imaging for environmental stress diagnosis in crops. *Sensors*, 24(5), p.1442.
- [5] Swaminathan, B. and Vairavasundaram, S., 2024. D2CNN: Double-staged deep CNN for stress identification and classification in cropping system. *Agricultural Systems*, 216, p.103886.
- [6] Sai, K., Sood, N. and Saini, I., 2024. Sorted sample-wavelet feature clustering based on the range for classification of multiple nutrient deficiencies in tomato plants. *Computers and Electronics in Agriculture*, 225, p.109263.
- [7] Meher, P.K., Sahu, T.K., Gupta, A., Kumar, A. and Rustgi, S., 2024. ASRpro: A machine - learning computational model for identifying proteins associated with multiple abiotic stress in plants. *The Plant Genome*, 17(1), p.e20259.
- [8] Bera, A., Bhattacharjee, D. and Krejcar, O., 2024. PND-Net: plant nutrition deficiency and disease classification using graph convolutional network. *Scientific Reports*, 14(1), p.15537.
- [9] Panahi, B., Ghavidel, B. and Shahgoli, P., 2024. Harnessing machine learning approach for phenotyping and deciphering the plants biotic and abiotic stresses responsive molecular mechanisms. *Crop Biotechnology*, 13(46), pp.75-90.
- [10] Cao, Y., Tian, D., Tang, Z., Liu, X., Hu, W., Zhang, Z. and Song, S., 2024. OPIA: an open archive of plant images and related phenotypic traits. *Nucleic Acids Research*, 52(D1), pp.D1530-D1537.
- [11] Blanc-Mathieu, R., Dumas, R., Turchi, L., Lucas, J. and Parcy, F., 2024. Plant-TFClass: a structural classification for plant transcription factors. *Trends in Plant Science*, 29(1), pp.40-51.
- [12] Demilie, W.B., 2024. Plant disease detection and classification techniques: a comparative study of the performances. *Journal of Big Data*, 11(1), p.5.
- [13] Kaur, S., Anandaram, H., Ahmad, A., Kumari, A., Bhosale, V.K., Joshi, K., Saini, D.J.B. and Krishna, G., 2024. A novel and optimized IoT-ML based plant classification, monitoring and prediction system. *International Journal of Information Technology*, pp.1-7.
- [14] Akbari, M., Sabouri, H., Sajadi, S.J., Yarahmadi, S. and Ahangar, L., 2024. Classification and prediction of drought and salinity stress tolerance in barley using GenPhenML. *Scientific Reports*, 14(1), p.17420.
- [15] Kukreja, V., Sharma, V. 2024, January. Elevating Crop Protection: Vision Transformer Models for Brassica Black Rot Detection. In 2024 International Conference on Intelligent and Innovative Technologies in Computing, Electrical and Electronics (IITCEE) (pp. 1-5). IEEE.
- [16] Kukreja, V., Sharma, V. and Sharma, R., 2024, January. Utilizing a Hybrid CNN-LSTM Model for Automated Severity Classification of Rice Panicle Blast Disease. In 2024 International Conference on Intelligent and Innovative Technologies in Computing, Electrical and Electronics (IITCEE) (pp. 1-5). IEEE.